Statistics

1. Population vs sample
2. Types of data
3. Descriptive statistics
   - measure of central tendency - mean, median, mode
   - measure of asymmetry - skewness and kurtosis
   - measure of variability - variance, standard deviation, coefficient of variation.
4. Inferential statistics
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   - z-test, z-table, z-score
   - t-test, t-table
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5. Advance statistics
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   - reject the null hypothesis
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6. Important stats concept to build ml model
   - mae (mean absolute error)
   - mse (mean squared error)

✓ rmse (root mean squared error)
anova
✓ ssr (sum of square regression)
✓ sst (sum of square total)
✓ sse (sum of square error)
r2
adjusted r2
regression table.

* Population VS Sample

Population is the complete group that we are interested in.

A sample is a small part taken from the population to represent it.

For eg: If the class has 60 students and I check the marks of 10 selected students then
60 students ≡ population
10 students ≡ sample

* Types of data

- Categorical (Classification)
- Numerical
  - Discrete (Clustering)
  - Continuous (Regression)

① Classification
Classification means putting data into predefined categories (classes). The outputs are labels

Eg: Spam / Not Spam
Pass / Fail
Yes / NO

② Regression
Means predicting a continuous numerical value. The output is a number

Eg: House Price Prediction
Salary Prediction.

③ Clustering:
Clustering means grouping similar data points together. No labels are given in advance.

Eg: Customer Groups.

* Descriptive Statistics

① Measures of central tendency: mean, median, mode.

① Mean: Mean is the sum of all values divided by total number of values

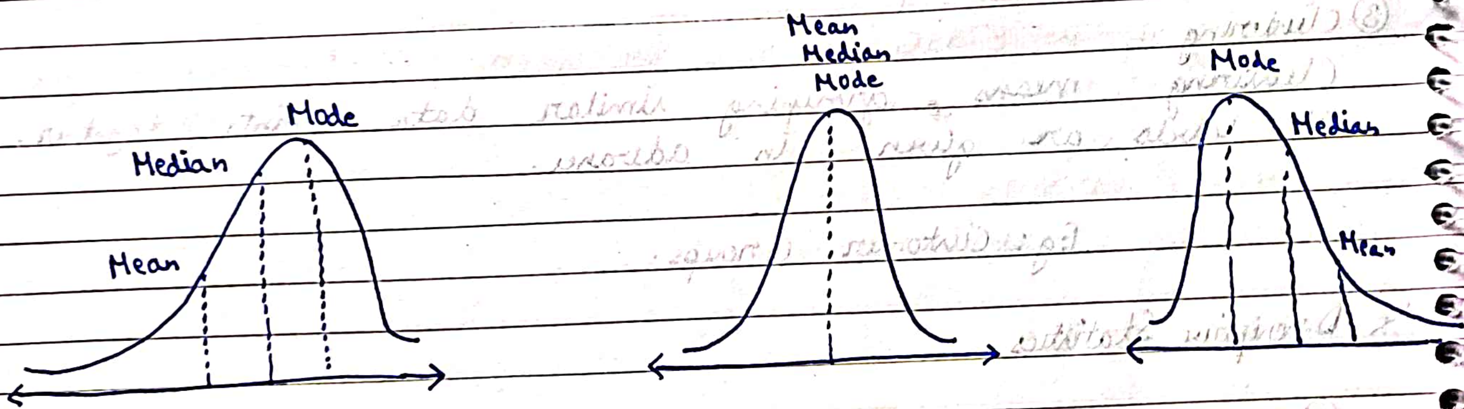
$$\text{Mean} = \frac{\text{Sum of all Values}}{\text{Number of values}}$$

② Median:
It is the middle value when data is arranged in order.

③ Mode
It is the value that appears most frequently.

② Measures of asymmetry - skewness and kurtosis.

① Skewness
Measures the asymmetry of the data distribution and helps us understand how values are spread around the mean.



Left Skewed
-ve Skewed

Mean < Median < Mode

No skew
Normal Distribution

Mean = Median = Mode

Right Skewed
+ve Skewed

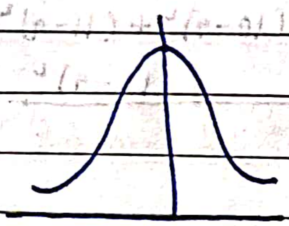
Mean > Median > Mode

② Kurtosis

Helps measures the peak or height of the data distribution and helps us understand the shape of frequency distribution.

Platykurtic

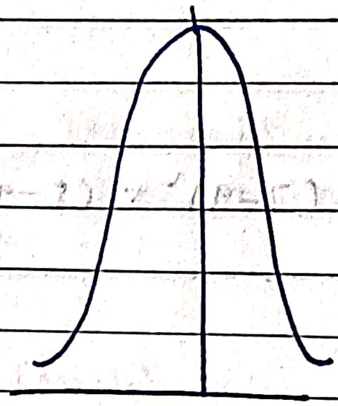
Kurtosis < 3



Mesokurtic

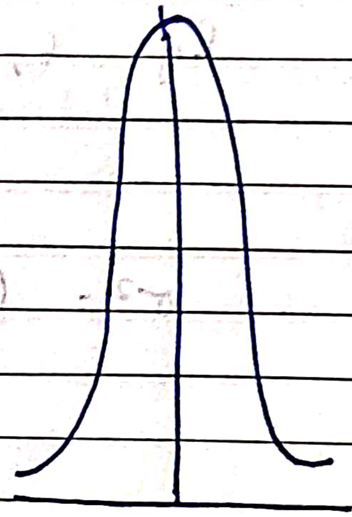
Normal Distribution

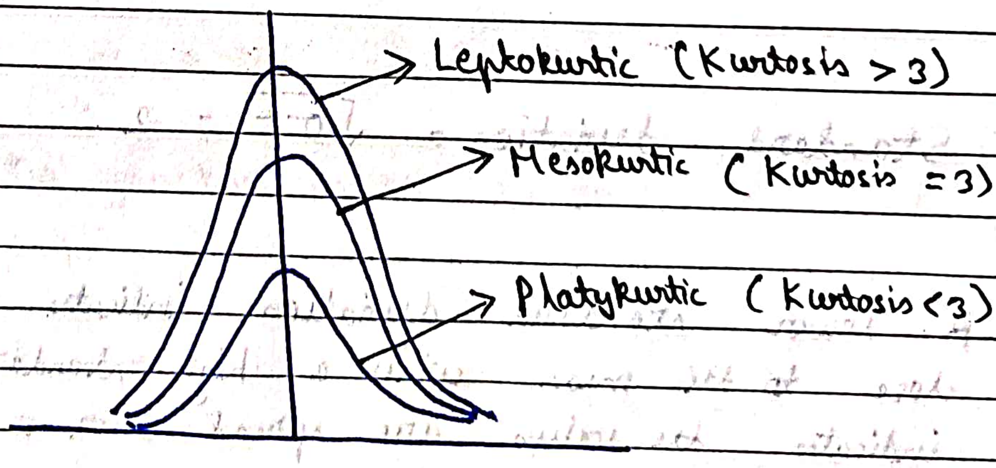
Kurtosis = 3



Leptokurtic

Kurtosis > 3





③ Standard deviation vs Variance

Variance is a measure of the spread or dispersion of a set of data points.

$$\text{Variance } \sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{N}$$

Eg: 6, 7, 8, 9, 10, 11, 12 $\quad \bar{x} = 9$

$$\sigma^2 = \frac{(6-9)^2 + (7-9)^2 + (8-9)^2 + (9-9)^2 + (10-9)^2 + (11-9)^2 + (12-9)^2}{7}$$

$$= \frac{9 + 4 + 1 + 0 + 1 + 4 + 9}{7} = 4$$

Standard deviation $= \sqrt{\sigma^2} = 2$

A lower standard deviation indicates the values tend to close to the mean while a high standard deviation indicates the values are spread over a wider range.

④ Coefficient of Variation:

It is a relative ~~variation~~ measure of dispersion. It is based on the mean and standard deviation of a frequency distribution.

$$C.V = \frac{\sqrt{\sigma^2}}{\bar{x}} \times 100$$

$$C.V = \frac{\text{Standard deviation}}{\text{Mean}} \times 100$$

More percentage = Less consistence
Less percentage = More consistence

* Inferential Statistics

We study a small group (sample) and infer about the big group (population)

Q When to use T-test vs z-test

Do you know the population σ

- Yes → Is the sample size above 30
  - → $\boxed{\text{Use z-test}}$
  - → $\boxed{\text{Use t-test}}$
- No → $\boxed{\text{Use t-test}}$

* Confidence Interval

A confidence interval gives a range of values within which the true population parameter is likely to lie. Instead of one value, we give a range.

$$\boxed{\text{Confidence Interval} = \text{Estimate} \pm \text{Margin of Error}}$$

* Advanced Statistics

① Hypothesis Testing

Hypothesis testing is a statistical method used to make a decision about a population using sample data.

In simple words to check whether a claim is true or false using data.

Hypothesis Testing

Null Hypothesis H_0
- Default assumption
- No change / no effect

Alternative Hypothesis. H_1
- Opposite of null hypothesis
- What we want to prove.

Level of significance (α)
- Probability of rejecting a true null hypothesis
- Common values : 0.05 (5%), 0.01

$\alpha = 0.05$ means 5% risk of wrong rejection.

Errors in Hypothesis Testing

① Type I Error (α error) Rejecting the true null hypothesis
Rejecting the true null hypothesis

Example:
Null Hypothesis (Ho):
The person is innocent

Type I Error:
We say guilty but the person is actually innocent

② Type II Error (β error) Accepting the false null hypothesis
Failing to reject the null hypothesis when it is actually false.

Example:
Null Hypothesis
The person is innocent

We say innocent but the person is actually guilty

⑥ Important stats concept to build ml model

① mae (mean absolute error)
Means how much our predictions are wrong. It tells us about the average absolute difference between actual values and predicted values.

$$MAE = \frac{1}{n} \sum |y_{actual} - y_{predicted}|$$

Actual	Predicted	Absolute Error
100	90	10
150	140	10
200	210	10

$$MAE = \frac{10+10+10}{3} = 10$$

② Mean Squared Error

The average of the squared differences b/w actual values and predicted values.

$$MSE = \frac{1}{n} \sum (y_{actual} - y_{predicted})^2$$

Actual	Predicted	Error	Squared error
50	45	5	25
60	70	-10	100
70	65	5	25

$$MSE = \frac{25+100+25}{3} = 50$$

③ Root Mean Squared Error

The square root of the average of squared differences between actual and predicted values

$$RMSE = \sqrt{\frac{1}{n} \sum (y_{actual} - y_{predicted})^2}$$

Actual	Predicted	Error	Squared Error
50	45	5	25
60	70	-10	100
70	65	5	25

$$MSE = \frac{25 + 100 + 25}{3} = 50$$

$$RMSE = \sqrt{50} \approx 7.07$$

④ SST - Total Sum of Squares
SST measures the total variation in the data.

It shows how much the actual values differ from the overall mean

$$SST = \Sigma (y_i - \bar{y})^2$$

⑤ SSR - Regression Sum of Squares

It shows how much the predicted values differ from the mean.

$$SSR = \Sigma (\hat{y}_i - \bar{y})^2$$

⑥ SSE - Error Sum of Squares.

It shows how much actual values differ from predicted values

$$SSE = \Sigma (y_i - \hat{y}_i)^2$$

$$SST = SSR + SSE$$

* R^2

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$